1. Network Implementation & Print Services

Network Implementation

Linux provides several tools and configuration files to implement and manage networking.

Key Networking Files:

File	Purpose
/etc/hostname	Sets system hostname
/etc/hosts	Local name resolution
/etc/network/interfaces	Debian-based network config
/etc/sysconfig/network-scripts/ifcfg-*	Red Hat-based config
/etc/resolv.conf	DNS servers

Key Commands:

bash

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```
ip a            # Show IP address ip r            # Show routing table ip link         # Show interfaces nmcli           # Manage NetworkManager ifconfig
# Legacy interface config ping, traceroute  # Connectivity tests
```

Network Security:

- Use **iptables** or **firewalld** for firewall rules
- **fail2ban** to block malicious IPs

Print Services

Linux supports print services via **CUPS** (Common Unix Printing System).

CUPS Features:

- Web interface: http://localhost:631
- Supports IPP (Internet Printing Protocol)
- Enables network printing via Samba, LPD, or IPP

Useful Print Commands:

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lp file.txt # Print file lpstat -p # Show printers lpq # View print queue cancel <job-id> # Cancel job lpadmin -p printer_name # Add/manage printer

Configuration File:

- /etc/cups/cupsd.conf Core CUPS config

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2. Service Management

Modern Linux systems use systemd

Basic Commands:

bash

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```
systemctl start apache2.service      # Start service systemctl stop apache2.service       # Stop service systemctl restart apache2.service    # Restart systemctl enable apache2           # Enable at boot systemctl disable apache2          # Disable at boot systemctl status apache2           # Check status
```

Older systems use init or service

bash

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service apache2 start chkconfig --list

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3. System Configuration Files

Common System Config Files:

File	Description
/etc/fstab	Mount filesystem automatically at boot
/etc/hostname	System's hostname
/etc/hosts	Static mapping of IP to hostname
/etc/passwd	User account info
/etc/group	Group account info
/etc/shadow	User passwords (hashed)

/etc/resolv.conf	DNS nameservers
/etc/network/interfaces or /etc/sysconfig/network-scripts	Interface configs
/etc/sysctl.conf	Kernel parameters
/etc/crontab	Scheduled jobs

Sysctl:

To modify kernel parameters at runtime:

bash

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sysctl -w net.ipv4.ip_forward=1

To persist:

bash

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```
echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf
```